

# ISMS V3 Octopus Cable Connection Key

| ISMS Housing Side |                 |                   | Exterior Connection Side                                                                                                                                                                                                                                                                                                        |                   |                                               |         |                 |                   |
|-------------------|-----------------|-------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-----------------------------------------------|---------|-----------------|-------------------|
| Pin No.           | Wire Base Color | Wire Stripe Color | Routed To                                                                                                                                                                                                                                                                                                                       | SubConn Connector | Pin Role                                      | Pin No. | Wire Base Color | Wire Stripe Color |
| 1                 | Black           |                   | Vehicle / Topside                                                                                                                                                                                                                                                                                                               | MCIL8M            | Main Power V+ 1 (VIN1)                        | 1       | Black           |                   |
| 2                 | White           |                   | Vehicle / Topside                                                                                                                                                                                                                                                                                                               | MCIL8M            | Main Power V+ 2 (VIN2)                        | 2       | White           |                   |
| 3                 | Red             |                   | Vehicle / Topside                                                                                                                                                                                                                                                                                                               | MCIL8M            | Main Power GND (VIN-)                         | 3       | Red             |                   |
| 4                 | Green           |                   | Vehicle / Topside                                                                                                                                                                                                                                                                                                               | MCIL8M            | tx/rx Arduino (tx)                            | 4       | Green           |                   |
| 5                 | Orange          |                   | Vehicle / Topside                                                                                                                                                                                                                                                                                                               | MCIL8M            | tx/rx Arduino (rx)                            | 5       | Orange          |                   |
| 6                 | Blue            |                   | Vehicle / Topside                                                                                                                                                                                                                                                                                                               | MCIL8M            | tx/rx RGA (tx)                                | 6       | Blue            |                   |
| 7                 | White           | Black             | Vehicle / Topside                                                                                                                                                                                                                                                                                                               | MCIL8M            | tx/rx RGA (rx)                                | 7       | White           | Black             |
| 8                 | Red             | Black             | Vehicle / Topside                                                                                                                                                                                                                                                                                                               | MCIL8M            | Communication GND (shared)                    | 8       | Red             | Black             |
| 9                 | Green           | Black             | Fluid Pump                                                                                                                                                                                                                                                                                                                      | MCIL5F            | Fluid Pump ■ GND (Shared)                     | 1       | Black           |                   |
| 10                | Orange          | Black             | Fluid Pump                                                                                                                                                                                                                                                                                                                      | MCIL5F            | Fluid Pump ■ Analog (0-5 V)                   | 2       | White           |                   |
| 11                | Blue            | Black             | Fluid Pump                                                                                                                                                                                                                                                                                                                      | MCIL5F            | Fluid Pump ■ Power (+12v)                     | 3       | Red             |                   |
| N/a               | N/a             | N/a               | Fluid Pump                                                                                                                                                                                                                                                                                                                      | MCIL5F            | UNUSED Fluid Pump ■ Tachometer                | 4       | Green           |                   |
| 12                | Black           | White             | Fluid Pump                                                                                                                                                                                                                                                                                                                      | MCIL5F            | Fluid Pump ■ Fwd/Rev 0/5v                     | 5       | Orange          |                   |
| 13                | Red             | White             | PH Probe                                                                                                                                                                                                                                                                                                                        | MCIL6F            | LX/TX: pH Power V- (GND)                      | 1       | Black           |                   |
| 14                | Green           | White             | PH Probe                                                                                                                                                                                                                                                                                                                        | MCIL6F            | LX: USB Data (-) TX: RS485 Data               | 2       | White           |                   |
| 15                | Blue            | White             | PH Probe                                                                                                                                                                                                                                                                                                                        | MCIL6F            | LX/TX: pH Power V+ (+5 VDC)                   | 3       | Red             |                   |
| 16                | Black           | Red               | PH Probe                                                                                                                                                                                                                                                                                                                        | MCIL6F            | LX:USB Data (+) TX: pH Analog Signal (0-3.3v) | 4       | Green           |                   |
| N/a               | N/a             | N/a               | PH Probe                                                                                                                                                                                                                                                                                                                        | MCIL6F            | UNUSED LX: Shield TX: RS485 Data              | 5       | Orange          |                   |
| N/a               | N/a             | N/a               | PH Probe                                                                                                                                                                                                                                                                                                                        | MCIL6F            | UNUSED pin for Compatibility                  | 6       | Blue            |                   |
| Compatibility     |                 |                   | The 8 pin vehicle/topside connector pinout is backwards compatible with vehicles that connected to prior ISMS versions. The 16 pin main bottle connector, fluid pump, and PH probe connectors ARE NOT backwards compatible.                                                                                                     |                   |                                               |         |                 |                   |
| Locking Sleeves   |                 |                   | All Subconn Male (MCILxM) connectors use a female locking sleeve. Conversely the Female Subconns (MCILxF) use a male locking sleeve. This gives the exposed pins a bit of protection.                                                                                                                                           |                   |                                               |         |                 |                   |
| pH Probe          |                 |                   | This cable is default compatible with pyroscience aquapHOx-LX model pH logger but can be adapted to fit aquapHOx-TX model pH sensors w an adapter cable and firmware changes. The pin role for the pH probe is specified based on how the pin would map to the connector on an aquapHOx-LX with “LX:” or aquapHOx-TX with “TX:” |                   |                                               |         |                 |                   |

### Topside / In Vehicle Connector Pinout

| SubConn Connector | Pin Role                                | SubConn Pin No. | Wire Base Color | Wire Stripe Color | Topside Interface Connector                                                        |
|-------------------|-----------------------------------------|-----------------|-----------------|-------------------|------------------------------------------------------------------------------------|
| MCIL8F            | Main Power V+ 1 (VIN1)                  | 1               | Black           |                   | Bare Lead or 2 pin connector (Microfit 3.0)                                        |
| MCIL8F            | Main Power V+ 2 (VIN2)                  | 2               | White           |                   | Bare Lead or 2 pin connector (Microfit 3.0)                                        |
| MCIL8F            | Main Power GND (VIN-)                   | 3               | Red             |                   | Bare Lead or 2 pin connector (Microfit 3.0)                                        |
| MCIL8F            | Arduino RX (Swapped from downstream TX) | 4               | Green           |                   | To DB9 Serial > RS232 to USB converter > Computer                                  |
| MCIL8F            | Arduino TX (swapped from downstream RX) | 5               | Orange          |                   | To DB9 Serial > RS232 to USB converter > Computer                                  |
| MCIL8F            | RGA RX (Swapped from downstream TX)     | 6               | Blue            |                   | To DB9 Serial with 'fake-handshake' null modem > RS232 to USB converter > Computer |
| MCIL8F            | RGA TX (Swapped from downstream RX)     | 7               | White           | Black             | To DB9 Serial with 'fake-handshake' null modem > RS232 to USB converter > Computer |
| MCIL8F            | Communication GND (shared)              | 8               | Red             | Black             | Bare Lead or 2 pin connector (Microfit 3.0)                                        |